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Lab Practical #06:

Study the application layer protocol DNS, DHCP, FTP.

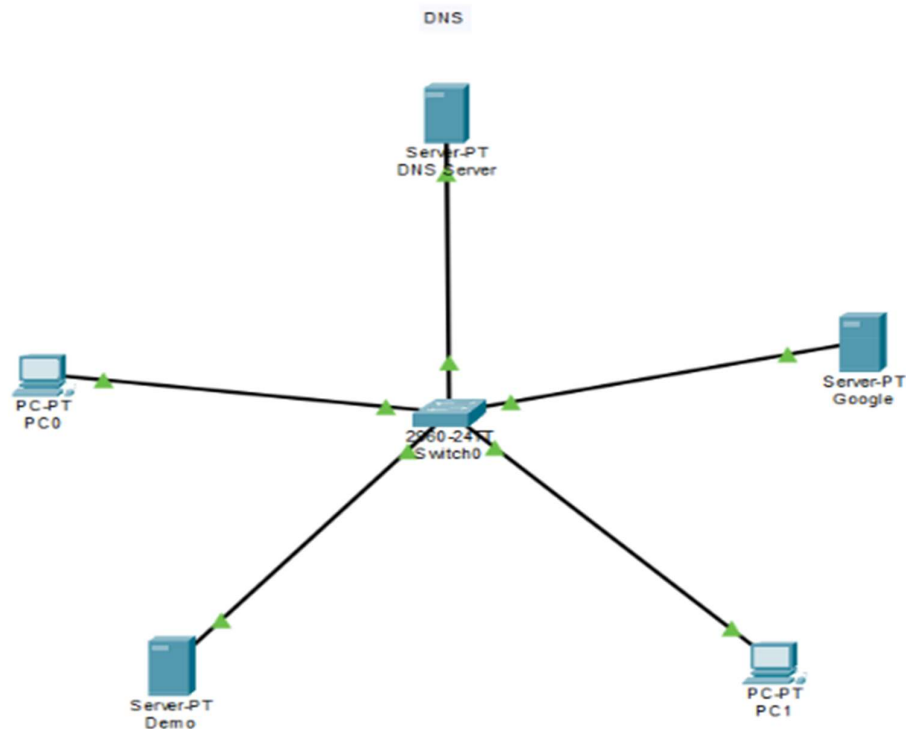
Practical Assignment #06:

1. Implement the application layer protocol DNS, DHCP, and FTP. Also check connectivity between them using ping command or PDU utility.

1. DNS

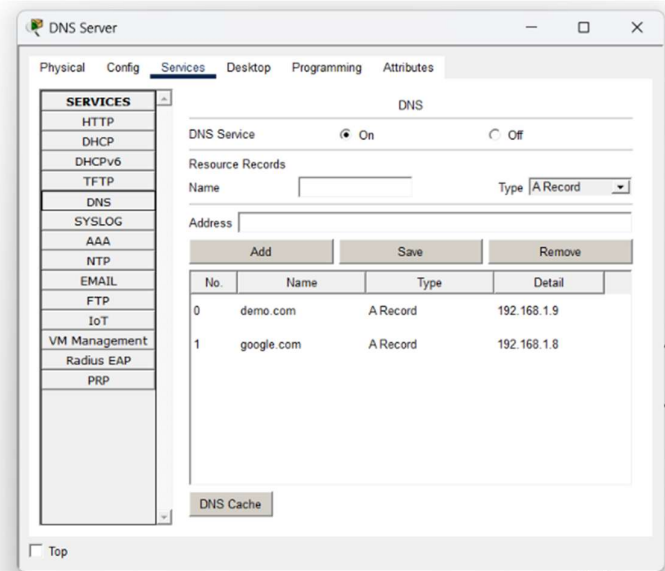
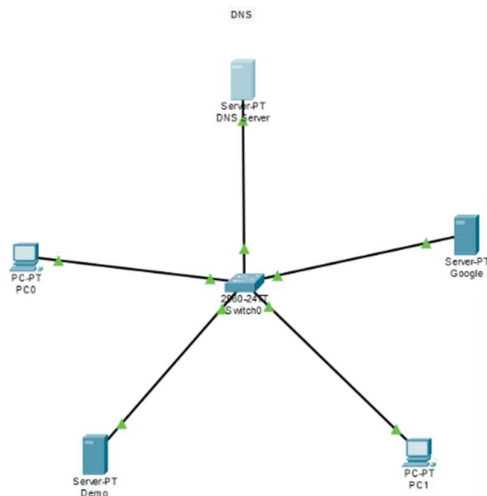
The DNS server maps domain names to IP addresses. The topology includes a DNS server, a Google Server, and a Demo Server.

Screenshot 1: Network Topology

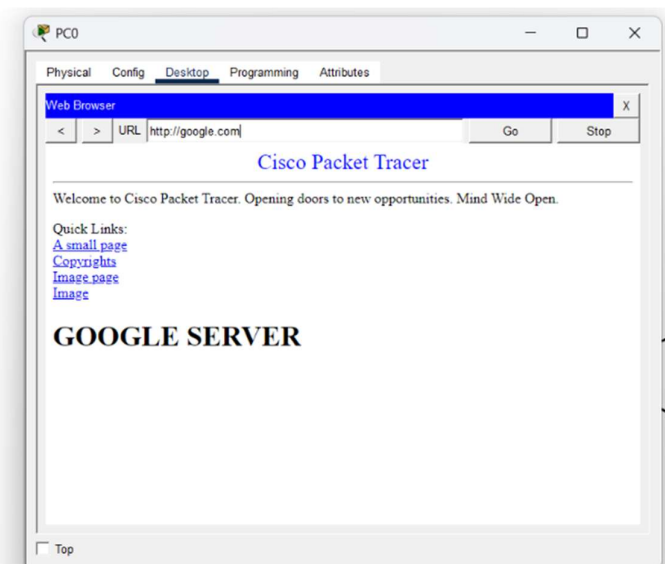
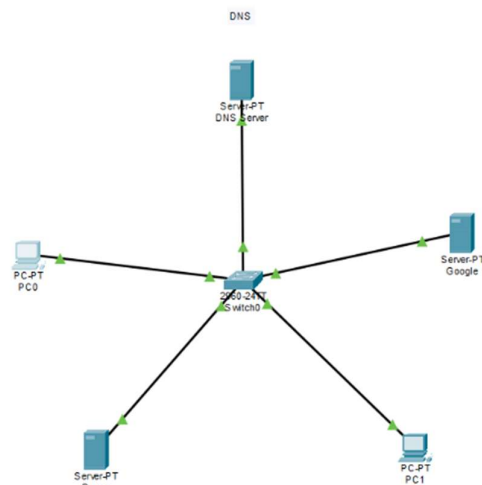


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Screenshot 2: DNS Server Configuration

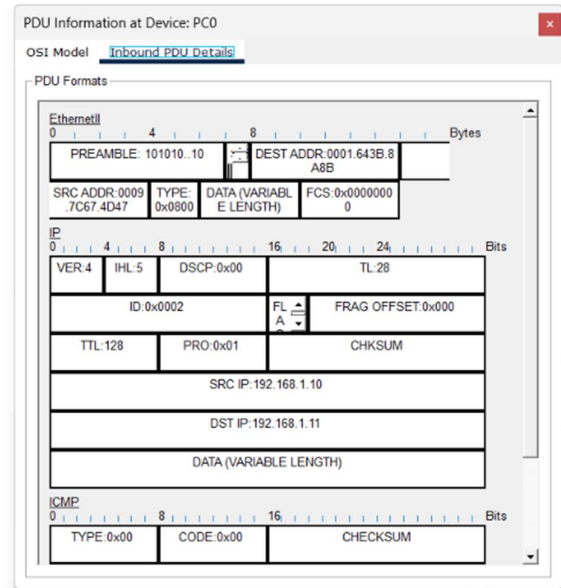
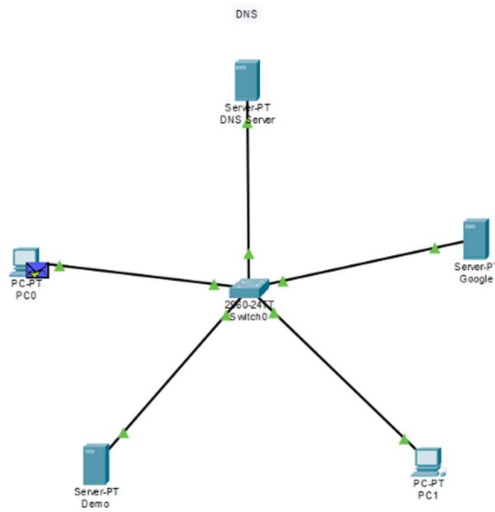


Screenshot 3: DNS Verification



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Screenshot 4: PDU



IP address Table

Device	IP Address	Subnet Mask
DNS Server	192.168.1.7	255.255.255.0
Google Server	192.168.1.8	255.255.255.0
Demo Server	192.168.1.9	255.255.255.0
PC0	192.168.1.11	255.255.255.0
PC1	192.168.1.10	255.255.255.0

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Procedure

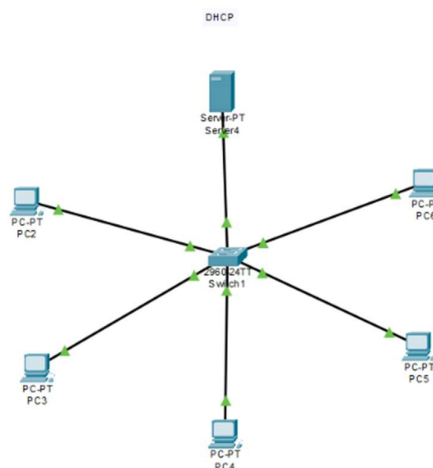
1. Enable the DNS service on the server.
2. Create a DNS record.
3. Enter:
 - Domain Name (e.g., www.server.com)
 - IP Address of the server.
4. Save the DNS record.
5. Configure the client PC to use the DNS server.
6. Open Command Prompt.
7. Ping the domain name or access it using the Web Browser.

Observation

The client successfully resolved the domain name into its IP address using the DNS server.

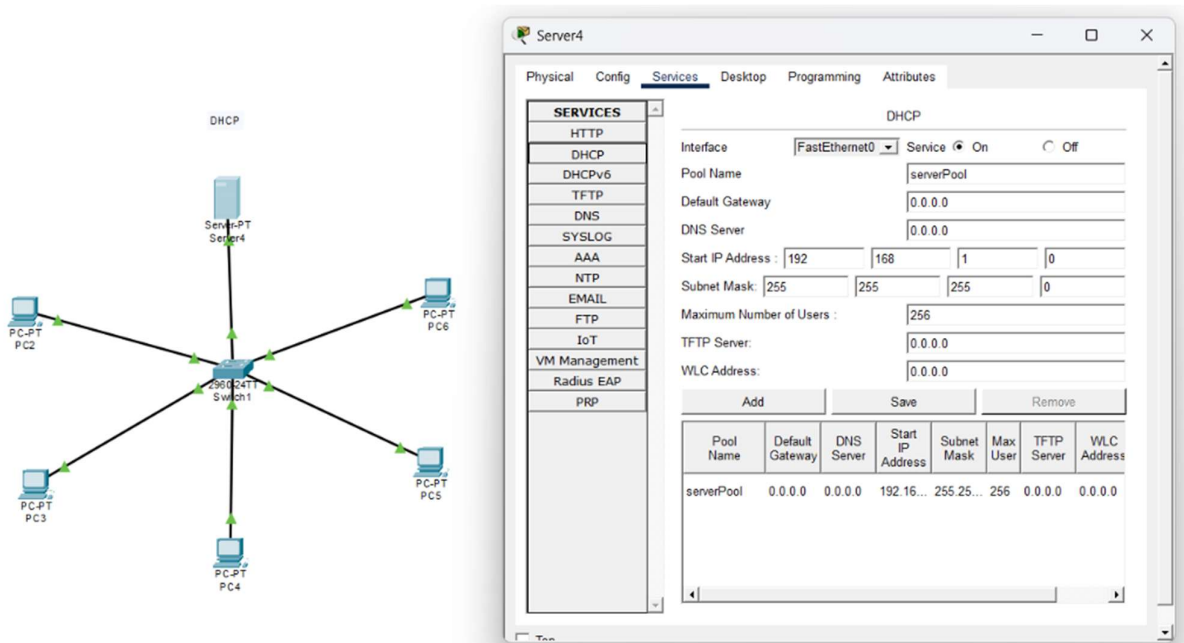
2. DHCP

Screenshot 1: Network Topology

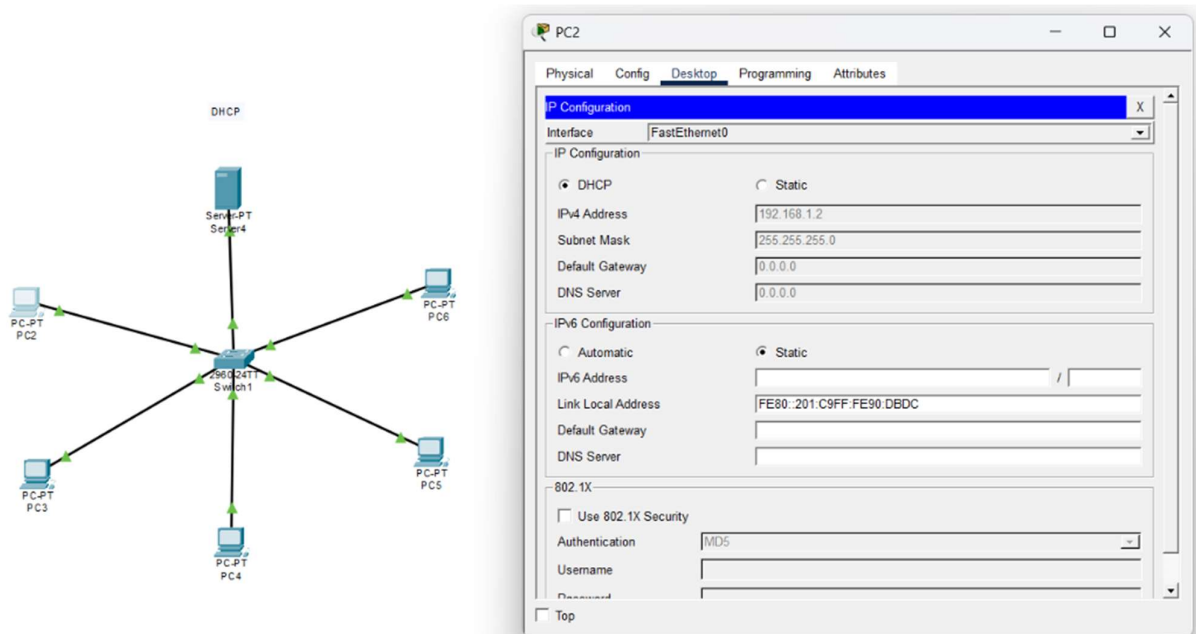


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Screenshot 2: DHCP Server Configuration

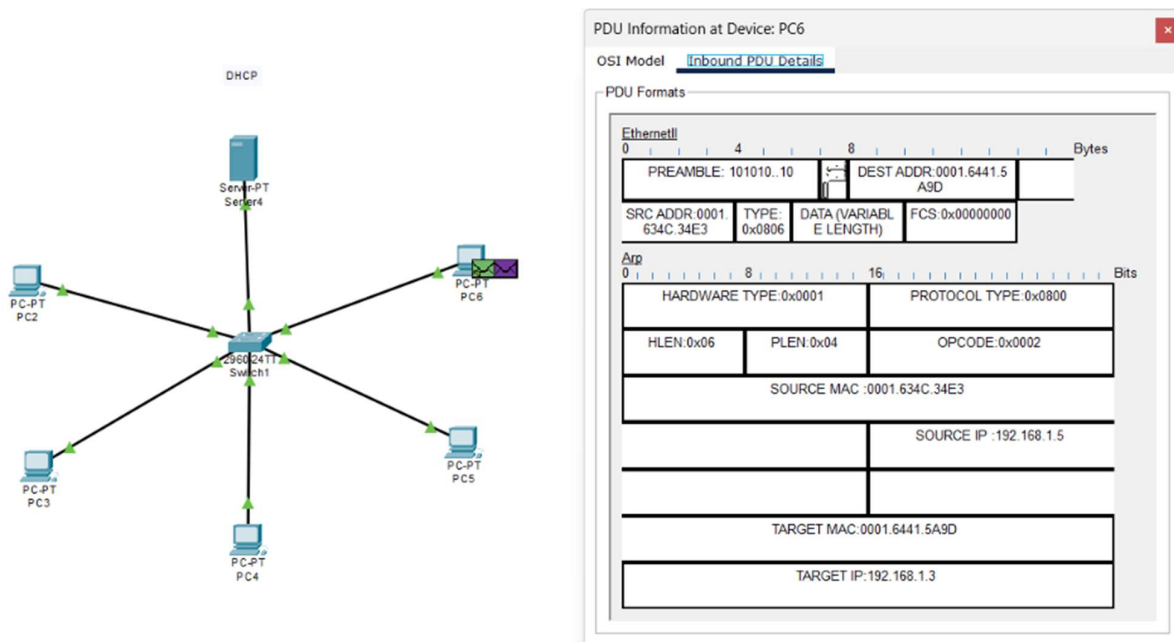


Screenshot 3: DHCP Verification



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Screenshot 4: PDU



IP address Table

Device	Assigned IP Address	Subnet Mask
DHCP Server	192.168.1.1	255.255.255.0
PC2	192.168.1.6	255.255.255.0
PC3	192.168.1.5	255.255.255.0
PC4	192.168.1.2	255.255.255.0
PC5	192.168.1.4	255.255.255.0
PC6	192.168.1.3	255.255.255.0

Procedure

1. Place one server, one switch, and five PCs in Cisco Packet Tracer.
2. Connect all devices using Copper Straight-Through cables.
3. Assign a static IP address to the server.
4. Enable the DHCP service on the server.
5. Create a DHCP pool.
6. Configure:

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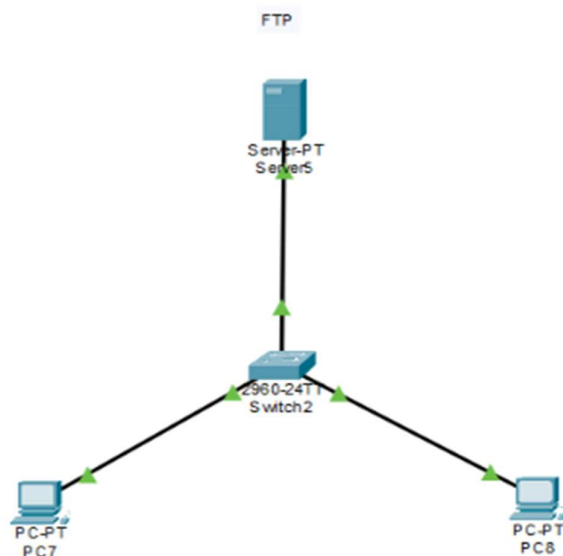
- Pool Name
 - Start IP Address
 - Subnet Mask
 - Maximum Users
7. Save the configuration.
 8. Configure all PCs to obtain IP addresses automatically.
 9. Verify that each PC receives an IP address from the DHCP server.

Observation

The DHCP server successfully assigned IP addresses automatically to all client computers.

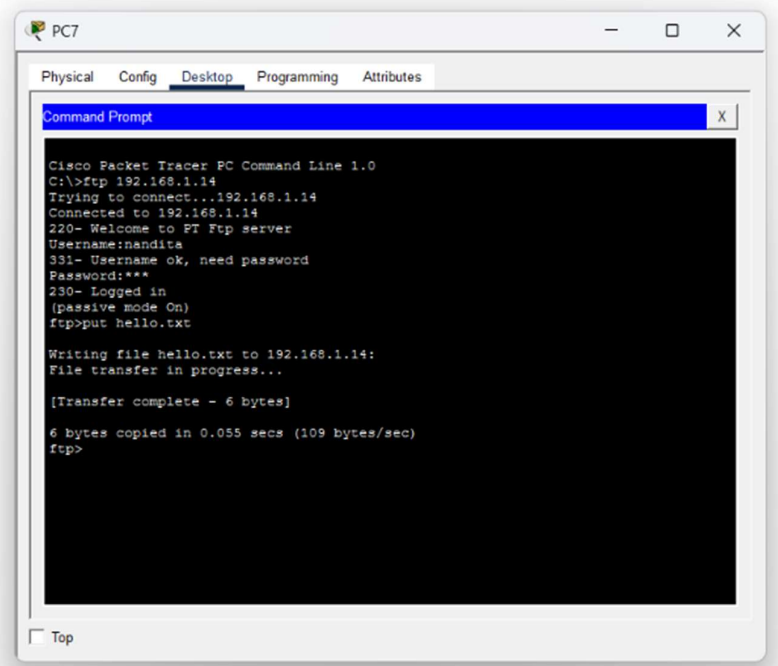
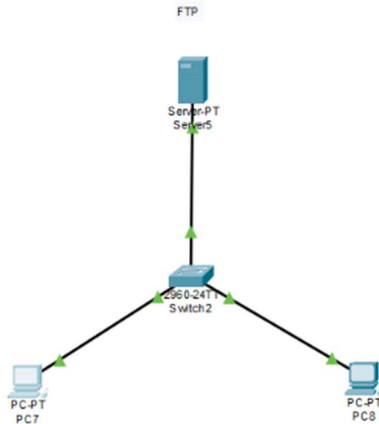
3. FTP

Screenshot 1: Network Topology



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Screenshot 2: FTP Server Configuration

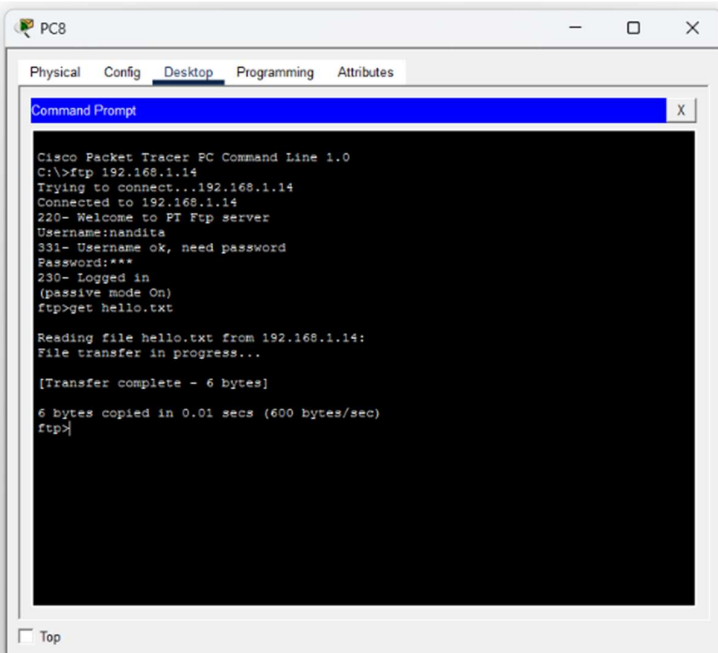
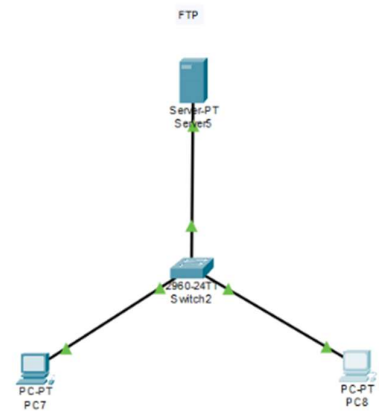


```
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 192.168.1.14
Trying to connect...192.168.1.14
Connected to 192.168.1.14
220- Welcome to PT Ftp server
Username:nandita
331- Username ok, need password
Password:***
230- Logged in
(passive mode On)
ftp>put hello.txt

Writing file hello.txt to 192.168.1.14:
File transfer in progress...

[Transfer complete - 6 bytes]
6 bytes copied in 0.055 secs (109 bytes/sec)
ftp>
```

Screenshot 3: FTP Verification



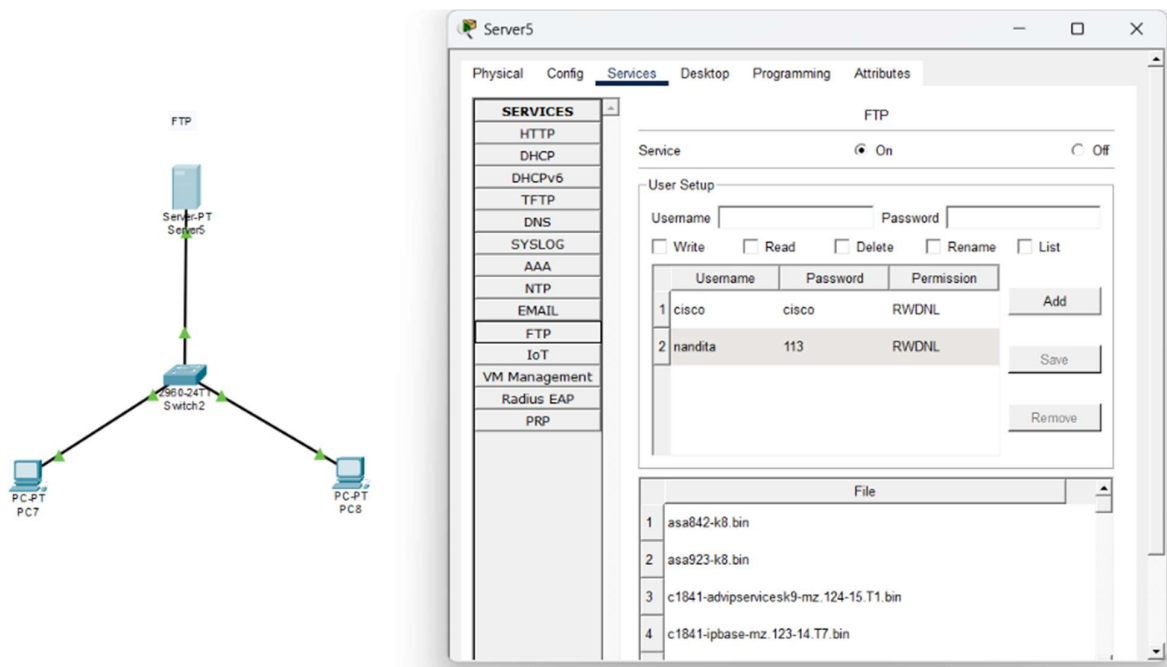
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220- Welcome to PT Ftp server
Username:nandita
331- Username ok, need password
Password:***
230- Logged in
(passive mode On)
ftp>get hello.txt

Reading file hello.txt from 192.168.1.14:
File transfer in progress...

[Transfer complete - 6 bytes]
6 bytes copied in 0.01 secs (600 bytes/sec)
ftp>
```

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Screenshot 4: PDU



IP address Table

Device	Assigned IP Address	Subnet Mask
FTP Server	192.168.1.14	255.255.255.0
PC7	192.168.1.12	255.255.255.0
PC8	192.168.1.13	255.255.255.0

Procedure

1. Enable the FTP service on the server.
2. Create a username and password.
3. Save the configuration.
4. Open Command Prompt on the client PC.
5. Connect to the FTP server using: ftp 192.168.1.1
6. Enter the username and password.



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7. Verify successful login and access to the FTP server.

Observation

The client successfully authenticated with the FTP server and established a file transfer session.

Real-Life Applications

DHCP (Dynamic Host Configuration Protocol)

- DHCP automatically assigns IP addresses and other network settings to devices connected to a network. It eliminates manual configuration and reduces configuration errors.
- **Example:**
When a smartphone or laptop connects to a home Wi-Fi router, the router automatically assigns an IP address using DHCP.

DNS (Domain Name System)

- DNS translates human-readable domain names into IP addresses, making it easy to access websites without remembering numerical IP addresses.
- **Example:**
Typing `www.google.com` into a web browser automatically resolves to Google's IP address through the DNS server.

FTP (File Transfer Protocol)

- FTP is used to upload and download files between a client and a server over a network.
- **Example:**
A web developer uploads website files from a computer to a web hosting server using FTP.